Yash Agarwal

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EDUCATION

Carnegie Mellon University, School of Computer Science — **Pittsburgh, PA**
Masters in Intelligent Information **Systems** - GPA - 3.9/4.0 — 2025 - 2026
- *Coursework - Parallel Computation, Computer Systems, DL Systems, LLM Systems, Deep Learning, Multimodal ML*
- *Advisor - Prof.* ***Tianqi Chen*** *(NVIDIA, creator of TVM, XGBoost)*

Indian Institute of Technology (IIT), Kharagpur — **Kharagpur, India**
Bachelors in **Computer Science** and Engineering - GPA 8.65/10 — 2019 - 2023
- **All India Rank 250** out of 1.2 million applicants for Joint Entrance Examination (JEE Advanced)
- *Publication* - YesBut: Evaluating Vision-Language Models — **EMNLP 2024**

WORK EXPERIENCE

Modular Inc — **San Francisco, CA**
AI Performance Engineering Intern – MAX Inference Engine — Summer 2026
- Shipped **DSpark speculative decoding** for **Gemma 4 12B/31B** reaching **519 tok/s**, block-parallel drafter on target **hidden states**, **NVFP4** targets, **FP8 KV cache**, **CUDA graph capture** - **4×** faster decode; beat **vLLM** TPOT by 28%
- Implemented **BLASST dynamic block-sparse attention** in the warp-specialized **FlashAttention-4** kernel on **Blackwell B200**, skipping near-zero KV blocks mid-**online softmax** - attention up to **42% faster**, **TTFT −20%** at preserved accuracy
- Fixed a serial softmax stall in the **FA4** kernel with **Nsight Compute** and per-warp **clock64** gantt timelines; shipped **cross-stage-P TMEM** scheduling with **QK-first MMA** ordering, freeing score buffers early - **3.4%** faster kernel
- Accelerated **pinned host-memory** allocation **10×** (50mins → 5mins) for tiered **KV-cache offload** via lock-free **NUMA-pinned** parallel page faulting at 99 GB/s pipelined into GPU registration, **18% faster than CUDA**'s VMM API
- Re-architected serving to batch prompt chunks into decode steps' idle FLOPs with speculation live, replacing exclusive **prefill** steps that froze every user 13–17 ms - prompt cost **53 → 16 µs/token**, per-token latency up to **12.9%** lower
- Built **model-free speculative decoding** from a **suffix tree** over generated tokens, tree ops injected as **CUDA host callbacks** inside the decode graph to preserve CPU-GPU **async scheduling** - **TPOT −21%** on code, **−11%** on **Spec-Bench**

Squarepoint Capital — **Bangalore, India**
Software Developer **C++** — 2023 - 2025
- Architected firm's first **FX Options** trading stack, built entire system in **C++20** and Python to pioneer new asset class
- Debugged critical production crashes using **gdb** core dump analysis and implemented **thread-safety** in memory access
- Redesigned **multithreaded** Python testing framework to single-threaded architecture eliminating **40%** test flakes

PROJECTS @ CARNEGIE MELLON

Needle - Optimizing CUDA GPU Kernels and Autodiff | DL Systems Course Project — **Fall 2025**
- Optimized Matmul **CUDA kernel** by tiling, vectorization and GMEM coalescing, achieving 85% performance of **cuBLAS**
- Implemented **Flash Attention**, including kernel fusion, tiling, online softmax, achieving 77% performance of PyTorch
- Architected automatic differentiation Tensor, complete with mathematical ops and nn.Modules for Conv, RNN, MHA
- Developed NDArray class supporting reshaping, broadcasting, summation, Matmul using **C++ and CUDA** backend

Malloc, cache, shell | Computer Systems Course Projects — **Spring 2026**
- Implemented **malloc** using **mmap** and segregated free-lists, optimizing using **Better Fit**, **miniblocks**, removing footers
- Implemented a **cache** simulator, optimized matrix transpose by tiling and eliminating **evictions**, 4th in a class of 200
- Parallelized a File System with **RW locks**, enabling **concurrent** reads and writes to shared inodes, blocks, directories

WhisQer - Quantization trained Conformer | Deep Learning Course Project — **Fall 2025**
- Architected **quantized** Conformer ASR with 2-bit and 1-bit precision variants, with learnable scaling parameters, compressing a 108MB FP32 model to 45.6MB 2-bit by quantizing encoder FFN and attention projections
- Implemented teacher student co-training combining CTC and KL divergence loss and straight through estimator

SKILLS

Languages - C++ 23, Python, Rust, Bash | **ML -** PyTorch, JAX, Numpy, CUDA, Triton, Mojo